Unit: Unit 4: Absolute Value **Name:** ____________
Topic: Absolute Value Function and its Graph **Date:** ________

Foundation (Jalapeno)

Q1. What is the vertex of the graph of $y = |x|$?
(A) (0, 0)
(B) (0, 1)
(C) (1, 0)
(D) (1, 1)
(E) (-1, 0)

Q2. What is the axis of symmetry of the graph of $y = |x - 2|$?
(A) x = -2
(B) x = 0
(C) x = 2
(D) x = 4
(E) y = 0

Q3. Which of the following functions has a graph that opens downwards?
(A) $y = |x|)$
(B) $y = -|x|)$
(C) $y = |x - 2|)$
(D) $y = |x| + 2)$
(E) $y = -|x| + 2)$

Q4. What is the equation of the graph of $y = |x|$ shifted 3 units to the right?
(A) $y = |x - 3|)$
(B) $y = |x + 3|)$
(C) $y = |x| + 3)$
(D) $y = |x| - 3)$
(E) $y = |x - 3| + 3)$

Q5. Which of the following is the graph of $y = |x| - 2$?
(A) A graph that is shifted 2 units down
(B) A graph that is shifted 2 units up
(C) A graph that is shifted 2 units to the right
(D) A graph that is shifted 2 units to the left
(E) A graph that is stretched vertically by a factor of 2

Q6. What is the equation of the graph of $y = |x|$ reflected across the x-axis?
(A) $y = |x|)$
(B) $y = -|x|)$
(C) $y = |x| + 1)$
(D) $y = |x| - 1)$
(E) $y = -|x| + 1)$

Q7. Which of the following functions has a graph that is the same as the graph of $y = |x|$
(A) $y = |x + 2|)$
(B) $y = |x - 2|)$
(C) $y = |x| + 2)$
(D) $y = |x| - 2)$
(E) $y = |x| + |x|)$

Q8. What is the equation of the graph of $y = |x|$ shifted 2 units up and 1 unit to the right?
(A) $y = |x - 1| + 2)$
(B) $y = |x + 1| + 2)$
(C) $y = |x| + 2)$
(D) $y = |x + 1| - 2)$
(E) $y = |x - 1| - 2)$

Open Ended Questions (FRQ)

Q9. Graph the function $y = |x - 2| - 1$ and identify the vertex, axis of symmetry, and direction of opening.

Q10. Find the equation of the graph of $y = |x|$ stretched vertically by a factor of 2 and shifted 1 unit to the left.

Chef's Tips

- Read each question carefully before answering
- Show all work for free response questions
- Check your answers before submitting your test